

Synopsis



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Relocation:	Yes
Visa Status:	US citizen
Notice:	1 month
Attributes:	20 years semiconductor processing / characterization 12 years data science / artificial intelligence 8 years pharmaceutical development
Education:	BS Chemical Engineering 1984 MS Material Science 1999 MS Physical Chemistry 2003

Overview

As we approach a new era of artificial general intelligence (AGI), where artificial intelligence approaches human aptitude, I plan to leverage my skills in computer programming, optical sensors, and process control...

When I first started out at MIT, my supervisor told me if you master the thin film deposition techniques, all the other steps in the semiconductor manufacturing process will be routine. At this point I had a *chemical engineering* degree, so I already had a reasonable grasp of the system processing aspects of semiconductor manufacturing. My next goal was to understand the energetics which were going on at the surface and the inherent microstructure which evolved through *materials science*. My final frontier in the triad was to explore the chemical mechanisms behind film growth, to determine what happened when the chemical species hit the surface and how the intermediates influence crystal quality through *physical chemistry*. With this set of tools, I have three different perspectives to address semiconductor growth issues.

With over 20 years as a Research Scientist, I have coauthored roughly 16 papers, half under the direction of John Haggerty at MIT, and the remainder with support from Klaus Bachmann at NC State. During the course of research, I have:

- (1) Invented a patentable 'symmetric proportional control' algorithm for laser cavity optimization, stabilizing the growth of thin films.
- (2) Fabricated the first laser-induced, chemical vapor deposition, amorphous silicon solar cell.
- (3) Developed a microwave plasma, chemical vapor deposition system, to create polycrystalline diamond from methane gas, in a regime where kinetics dominates over thermodynamics.
- (4) Monitored the surface evolution of compound semiconductor heterostructure films, in a chemical beam epitaxy system, with plane polarized reflectance spectroscopy, pioneered by our research group.

As we approach a new era of artificial general intelligence (AGI), where artificial intelligence approaches human aptitude, I plan to leverage my skills in computer programming, optical sensors, and process control to turbocharge semiconductor manufacturing, and lead the way toward quantum processor architecture. I invite you to visit my personal website to view current literature and computational projects.

Christopher J Harris

Profile

Chemical Engineer seeking a Process Engineer, Research & Development, or Data Science role, applying my core skills in:

chemical vapor deposition	plasma sources	x-ray diffraction
molecular beam epitaxy	laser excitation	electron microscopy
reactive ion etching	ellipsometry	python language
photolithography	cyclic voltammetry	statistical analysis
metallization techniques	heteroepitaxial structures	process control

Thesis

Real Time Reflectometry of Ga-based Compound Semiconductor Films on Silicon during Plasma Enhanced Molecular Beam Epitaxy, NCSU Materials Science Dept: **1999**.

Clifton Strengths**Character**

<i>Strategic</i>	faced with any given scenario, can quickly spot the relevant patterns.
<i>Learner</i>	have a great desire to learn and want to continuously improve.
<i>Ideation</i>	able to find connections between seemingly disparate phenomena.
<i>Futuristic</i>	inspired by the future and what could be.
<i>Self-Assurance</i>	possess an inner compass to instill confidence in decision making.

Experience

Prompt Engineer, Outlier AI: San Francisco, CA (10/24 to present)

- ▶ Generate prompt questions to evaluate AI model responses to python programming tasks.
- ▶ Write and rewrite AI model responses to prompt questions with accurate python code.

Network Member, Gerson Lehrman Group: Austin, TX (4/20 to present)

- ▶ Appear as an expert witness in a patent lawsuit regarding diamond thin films.
- ▶ Provide scientific insight on a myriad of semiconductor issues.

Quantum Trader, Independent (5/84 to present)

- ▶ Broaden the scope to include both fundamental evaluation and technical analysis.
- ▶ Combine statistics with digital signal processing to produce indicators with better trade signal clarity.
- ▶ Implement python programming to learn hedge fund strategies.
- ▶ Evaluate fundamental aspects of technology and pharmaceutical sectors.
- ▶ Deploy venture capital principles to choose lucrative issues, including initial public offerings.

Laboratory Technician, Genesis Biotechnology Group: Hamilton, NJ (7/20 to 10/21)

- ▶ Process COVID-19 nasal swabs in Biological Safety Level 2 hoods under CDC guidelines.
- ▶ Extract nucleic acid samples from blood, urine, spinal, and other body fluids using Vacuum Filtration or Magnetic Bead separation.
- ▶ Perform static Polymerase Chain Reaction (PCR) and dynamic PCR (qPCR).
- ▶ Provide in-house quality control, identifying errors in: sample placement, robot operation, and SOPs.
- ▶ Apply advanced process engineering techniques to make Biomek robot extractions more reliable.
- ▶ Modify machine code to customize Biomek robot extractions.

Research Assistant, Maine Chemistry Dept: Orono, ME (8/03 to 5/06)

- ▶ Synthesize aurophilic gold compounds, a potential drug candidate for arthritis.
- ▶ Apply cyclic voltammetry to investigate catalytic activity in gold compounds.
- ▶ Induce electrochemiluminescence in a ruthenium compound for DNA analysis.

Teaching Assistant, Rutgers Chemistry Dept: New Brunswick, NJ (1/00 to 1/03)

- ▶ Present lab techniques to students on all undergraduate levels, including PChem.
- ▶ Pass the cumulative exam, the written portion of a PhD degree.

Research Assistant, NCSU Materials Science Dept: Raleigh, NC (1/87 to 12/99)

- ▶ Develop a microwave plasma, chemical vapor deposition system, to create polycrystalline diamond from methane gas, in a regime where kinetics dominates over thermodynamics.
- ▶ Achieve unique ellipsoidal plasma, energizing the surface, relative to spherical geometry.
- ▶ Design a radio frequency nitrogen plasma source for GaN film growth.
- ▶ Monitor the surface evolution of compound semiconductor heterostructure films, in a chemical beam epitaxy system, with plane polarized reflectance spectroscopy.
- ▶ Derive substrate temperature from plane polarized reflectance intensity.

Research Specialist, MIT Advanced Energy Materials Lab: Cambridge, MA (11/84 to 1/87)

- ▶ Invent a new approach for process control to optimize laser power.
- ▶ Write a Pascal based data acquisition program for DOS environment in 1986, long before LabView enters the Windows market.
- ▶ Analyze optical signals from a ceramic powder reaction chamber, leading to a computer monitoring scheme, which replaces a human operator.
- ▶ Construct interferometer to measure film thickness, providing a realtime signal, to calibrate growth rate.
- ▶ Refine process control loop to stabilize laser power, producing a steady deposition rate with reliable material properties.
- ▶ Collect in-situ stress measurements of growing films, through deflection of an optical laser, as sample curvature evolves.
- ▶ Grow the first laser-induced, chemical vapor deposition, amorphous silicon solar cell.

Education

MS <i>Physical Chemistry</i>	Rutgers: New Brunswick, NJ	Jan 2003
MS <i>Material Science</i>	North Carolina State: Raleigh, NC	unofficial
BS <i>Chemical Engineering</i>	Texas A&M: College Station, TX	May 1984
HS <i>Diploma</i>	Waltham High: Waltham, MA	Jun 1979

Honor

Bausch & Lomb Science Award